

POSTGRADUATE DIPLOMA IN ICT

Capstone Project — Design Phase Submission

Project Title	[Enter full project title]
Student Full Name	[Enter full name]
Student Number	[Enter student number]
Supervisor Name	[Enter supervisor's full name and title]
Stream Category	[Select ONE: Application Development Computer Networks AI and Data Science Web Technology Information Systems]
Academic Year	[e.g. 2025 / 2026]
Submission Date	[DD Month YYYY]

Declaration

I declare that this submission is my own original work and has been prepared in accordance with the academic integrity policies of the institution.

Student Signature:

Date: _____

Table of Contents

1. Introduction

This section contextualises the project problem, defines the scope, and states the objectives. Be precise and concise — aim for 400–600 words.

1.1 Project Problem Statement

Describe the problem or challenge your project addresses. Explain why this problem is significant, who is affected, and what the implications of inaction are.

[State the problem your project addresses and its real-world significance]

1.2 Project Scope

Define what is included and excluded from this project. Identify any system boundaries, user groups, geographic limitations, or technical constraints that apply.

[Define the boundaries of the project — what is in scope and what is explicitly excluded]

1.3 Project Objectives

List three to five measurable objectives that this project aims to achieve. Objectives should be specific, achievable, and aligned with the project problem.

[Objective 1: [State objective]]

[Objective 2: [State objective]]

[Objective 3: [State objective]]

[Objective 4 (if applicable): [State objective]]

[Objective 5 (if applicable): [State objective]]

2. Design Overview

Provide a high-level summary of your design approach. Introduce the system, network, model, or application architecture before the detailed sections that follow.

2.1 Summary of Design Approach

Summarise the overall design strategy. Identify the major components or subsystems and explain how they work together to address the problem.

[Provide a 200–300 word narrative overview of your design]

2.2 Justification of Chosen Approach

Explain why this particular design approach was selected over alternatives. Reference relevant design principles, patterns, or prior literature where appropriate.

[Justify your design approach, referencing alternatives considered and reasons for your selection]

3. Design Documents [30 Marks]

This section carries 30 marks. It must include both a High-Level Design document and a Detailed Design document. The depth of technical content should reflect the complexity of your project and stream.

3.1 High-Level Design (HLD)

Present an abstract, system-wide view of your design. Focus on architecture, component roles, and the relationships between subsystems. This should be understandable by a non-technical stakeholder.

3.1.1 System / Solution Architecture

Describe the major components or tiers of the solution (e.g. client layer, application layer, database layer, API layer). Reference your architecture diagram from Section 5.

[Provide a 200–400 word description of the architecture]

3.1.2 Component Roles and Interactions

For each major component identified above, describe its role and how it communicates with other components. Use a table if more than four components are present.

[Describe roles of each architectural component and their interaction mechanisms]

3.1.3 Technology Stack Summary

List the primary technologies, frameworks, platforms, and tools selected for implementation. Provide brief justification for each.

Layer / Component	Technology / Tool	Justification
Frontend / Presentation	<i>[Specify tool]</i>	<i>[Justify selection]</i>
Backend / Logic	<i>[Specify tool]</i>	<i>[Justify selection]</i>
Database / Storage	<i>[Specify tool]</i>	<i>[Justify selection]</i>
Infrastructure / Hosting	<i>[Specify tool]</i>	<i>[Justify selection]</i>
Other	<i>[Specify tool]</i>	<i>[Justify selection]</i>

3.2 Detailed Design (DDD)

Expand on the HLD by providing technical depth. This section must include technical specifications appropriate to your stream (see Section 8 for stream-specific guidance).

3.2.1 Module / Component Specifications

For each module or component, describe its inputs, outputs, processing logic, and dependencies. Use pseudocode, structured English, or decision tables where appropriate.

[Describe Module 1: Name, Input, Processing, Output, Dependencies]

[Describe Module 2: Name, Input, Processing, Output, Dependencies]

[Add additional modules as required]

3.2.2 Algorithms and Logic

Where your design includes non-trivial algorithms or business rules, describe the logic in pseudocode, flowchart, or structured English. Reference the relevant diagram in Section 5.

[Describe key algorithm(s) or business rules with pseudocode or structured notation]

3.2.3 Interface Specifications

Document the interfaces between components. For APIs, list endpoints; for UI, describe major screens; for networks, describe port/protocol mappings.

[Provide interface specifications relevant to your stream]

4. Diagrams and Visual Aids [25 Marks]

This section carries 25 marks. All diagrams must be clearly labelled, include a figure number and caption, and be referenced in the body text above. Diagrams must be of sufficient resolution for assessment. Include at least three distinct diagram types from the options below — your stream may require specific diagram types (see Section 8).

4.1 System / Architecture / Topology Diagram

Insert a high-level diagram showing the overall structure of your system, network, or solution. This may be a system context diagram, network topology diagram, or model pipeline overview depending on your stream.

[Figure 1: [Title of Diagram] — Insert diagram image here]

[Figure 1 Description: Explain what the diagram shows, the relationships depicted, and any design decisions it communicates]

4.2 Flowchart / Process Flow / Model Flow Diagram

Insert a flowchart illustrating a key process, user journey, algorithmic flow, or data transformation pathway. Use standard flowchart notation (ISO 5807 or equivalent).

[Figure 2: [Title of Diagram] — Insert diagram image here]

[Figure 2 Description: Describe the process flow, decision points, and any exception paths shown]

4.3 Sequence / Interaction Diagram

Insert a UML sequence diagram or interaction diagram that shows the chronological order of messages between system actors and components for a primary use case.

[Figure 3: [Title of Diagram] — Insert diagram image here]

[Figure 3 Description: Identify the actors, the use case depicted, and the sequence of interactions]

4.4 Data / Entity-Relationship / Data Flow Diagram

Insert an ERD, DFD (Level 0 and Level 1), or data schema diagram as appropriate to your stream. Ensure all entities, attributes, and relationships are labelled.

[Figure 4: [Title of Diagram] — Insert diagram image here]

[Figure 4 Description: Describe the data structures, entities, and relationships or data flows depicted]

4.5 Additional Diagrams (if applicable)

Include any additional diagrams required by your stream or your supervisor — for example, use-case diagrams, state machine diagrams, wireframes, or deployment diagrams.

[Figure 5: [Title of Diagram] — Insert diagram image here]

[Figure 5 Description: Explain relevance of this diagram to your design]

5. Prototypes and Mockups [25 Marks]

This section carries 25 marks. Present evidence of prototyping activity appropriate to your stream. Prototypes should demonstrate functional or visual progress toward the implementation. Include annotated screenshots, simulation outputs, or interactive prototype links where applicable.

5.1 Prototype Description

Describe the nature of your prototype: what it demonstrates, the scope of functionality included, and the tools or environment used to create it.

[Describe the prototype — what it represents, its scope, and the tools used to build it]

5.2 Prototype Screenshots / Simulation Outputs

Insert annotated screenshots of the prototype, UI mockups, or simulation output screens. Each image must be captioned and referenced. Annotations should highlight key design elements.

[Screenshot/Output 1: [Caption] — Insert annotated image here]

[Screenshot/Output 1 Description: Describe what is shown and what design decision it validates]

[Screenshot/Output 2: [Caption] — Insert annotated image here]

[Screenshot/Output 2 Description: Describe what is shown and what design decision it validates]

[Screenshot/Output 3: [Caption] — Insert annotated image here (add further screenshots as required)]

5.3 Explanation of Prototype Functionality

Explain how the prototype functions. Describe the user interactions or simulated behaviours that demonstrate alignment with the design requirements and project objectives.

[Describe how the prototype operates and which design decisions it validates]

5.4 Prototype Limitations and Next Steps

Identify any features or functionalities not yet implemented in the prototype and describe how these will be addressed in the implementation phase.

[Identify prototype limitations and planned enhancements during implementation]

6. Requirements Traceability [20 Marks]

This section carries 20 marks. Complete the traceability matrix below by linking each functional and non-functional requirement from the Analysis Phase to a specific design element in this document. Every row must be completed. Add additional rows if your project has more than five requirements.

Each requirement must be traceable to a specific design artefact (diagram, module, database table, API endpoint, etc.) with a clear justification explaining how the design element addresses the requirement.

Requirement ID & Description	Design Element / Artefact	Justification
[REQ-001: Provide requirement description]	[Specify design element or artefact]	[Explain how design addresses this requirement]
[REQ-002: Provide requirement description]	[Specify design element or artefact]	[Explain how design addresses this requirement]
[REQ-003: Provide requirement description]	[Specify design element or artefact]	[Explain how design addresses this requirement]
[REQ-004: Provide requirement description]	[Specify design element or artefact]	[Explain how design addresses this requirement]
[REQ-005: Provide requirement description]	[Specify design element or artefact]	[Explain how design addresses this requirement]

Note: Requirements should originate from the Analysis Phase document. Use consistent requirement IDs across all phase submissions.

8. Stream-Specific Design Requirements

Complete ONLY the subsection that corresponds to your selected stream. Delete or clearly mark the remaining subsections as N/A.

8.1 Application Development

Complete this section if your stream is Application Development.

8.1.1 User Interface Design

Provide annotated UI mockups or wireframes for all primary screens. Describe the design rationale for layout, colour scheme, and navigation structure.

[Insert UI mockup / wireframe images or attach as Appendix A]

[Describe UI design rationale and accessibility considerations]

8.1.2 Entity-Relationship Diagram (ERD)

Present a normalised ERD for your application's data model. Describe entities, attributes, primary keys, and relationships.

[Insert ERD diagram or reference Appendix B]

[Explain entity relationships and cardinality decisions]

8.1.3 Module Decomposition

List and describe the functional modules of the application, their responsibilities, and how they interact with one another.

[Provide a module hierarchy diagram or table]

8.2 Computer Networks

Complete this section if your stream is Computer Networks.

8.2.1 Network Topology Design

Present a clearly labelled network topology diagram. Identify all nodes, links, subnets, and hardware components. Justify the chosen topology.

[Insert network topology diagram or reference Appendix A]

[Justify topology choice (e.g. star, mesh, hybrid) with respect to requirements]

8.2.2 Protocol Specification

Document the protocols employed at each OSI layer relevant to your project. Include rationale for protocol selection.

[Provide a protocol table or narrative description per layer]

8.2.3 Simulation Environment

Describe the simulation tool used (e.g. Cisco Packet Tracer, GNS3). Present simulation outputs such as ping results, traffic logs, or performance graphs.

[Insert simulation screenshots or output logs]

8.3 AI and Data Science

Complete this section if your stream is AI and Data Science.

8.3.1 Model Architecture

Describe the chosen model type (e.g. CNN, regression, clustering). Justify the architectural decisions, including layers, hyperparameters, and training strategy.

[Provide a model architecture diagram or textual specification]

8.3.2 Data Pipeline

Describe the end-to-end data pipeline: acquisition, cleaning, feature engineering, splitting, and integration with the model. Reference tools and libraries used.

[Insert a data flow / pipeline diagram]

[Describe each pipeline stage with transformations applied]

8.3.3 Data Visualisation Mockups

Present mock outputs of dashboards or visualisations that will convey model results to end users or stakeholders.

[Insert visualisation mockups or dashboard wireframes]

8.4 Web Technology

Complete this section if your stream is Web Technology.

8.4.1 Wireframes and User Flow

Provide wireframes for all primary web pages and a user flow diagram illustrating how users navigate between pages.

[Insert wireframes (per page) — reference Appendix A]

[Insert user flow diagram]

8.4.2 API Specification

Document the RESTful or GraphQL API endpoints, including HTTP methods, request parameters, response structures, and error codes.

[Provide API specification table or reference an exported Postman / Swagger document]

8.4.3 Technology Stack Justification

Specify the front-end framework, back-end runtime, database, and hosting platform. Justify each technology choice with respect to project requirements.

[Provide a technology stack diagram or table]

8.5 Information Systems

Complete this section if your stream is Information Systems.

8.5.1 System Architecture Blueprint

Present a comprehensive architecture diagram showing system layers (presentation, business logic, data), integration points, and external systems.

[Insert system architecture diagram or reference Appendix A]

8.5.2 Database Schema

Provide a normalised database schema including all tables, columns, data types, primary keys, foreign keys, and relationships. Minimum Third Normal Form (3NF) required.

[Insert database schema diagram or ER diagram]

[Provide a data dictionary table (Field | Type | Description | Constraints)]

8.5.3 System Interaction Diagrams

Present sequence or collaboration diagrams illustrating how system components and external actors interact for at least three primary use cases.

[Insert interaction / sequence diagrams]

9. Evaluation of Design

This section demonstrates academic rigour by critically evaluating the quality and robustness of your design. Address each subsection thoroughly.

9.1 Feasibility Assessment

Evaluate the technical, operational, and economic feasibility of your design. Identify any feasibility risks and describe how they are mitigated.

[Discuss technical feasibility: Does the technology stack support the required functionality?]

[Discuss operational feasibility: Can the intended users realistically operate the system?]

[Discuss economic feasibility: Are resources (time, tools, cost) within acceptable limits?]

9.2 Scalability Considerations

Describe how the design could accommodate growth in users, data volume, or network traffic. Identify any architectural decisions made specifically to support scalability.

[Discuss horizontal or vertical scalability of your solution and the design decisions that support it]

9.3 Security Considerations

Identify the primary security risks relevant to your project (e.g. SQL injection, unauthorised access, data interception) and describe the design-level controls applied to mitigate them.

Security Risk	Likelihood (H/M/L)	Design-Level Mitigation
<i>[Identify risk]</i>	<i>[H / M / L]</i>	<i>[Describe mitigation]</i>
<i>[Identify risk]</i>	<i>[H / M / L]</i>	<i>[Describe mitigation]</i>
<i>[Identify risk]</i>	<i>[H / M / L]</i>	<i>[Describe mitigation]</i>

9.4 Risks and Mitigation

Identify up to five project risks at this design stage. For each risk, provide a likelihood rating, impact assessment, and mitigation strategy.

Risk Description	Likelihood	Impact	Mitigation Strategy
<i>[Describe risk]</i>	<i>[H/M/L]</i>	<i>[H/M/L]</i>	<i>[Describe mitigation]</i>
<i>[Describe risk]</i>	<i>[H/M/L]</i>	<i>[H/M/L]</i>	<i>[Describe mitigation]</i>
<i>[Describe risk]</i>	<i>[H/M/L]</i>	<i>[H/M/L]</i>	<i>[Describe mitigation]</i>
<i>[Describe risk]</i>	<i>[H/M/L]</i>	<i>[H/M/L]</i>	<i>[Describe mitigation]</i>

10. Conclusion

Provide a concise conclusion (200–350 words) that summarises the design phase outcomes.

10.1 Summary of Design Readiness

Summarise the key design decisions made during this phase. Confirm that all required design artefacts have been produced and that the design is ready to support implementation.

[Summarise the key design decisions and confirm design completeness]

10.2 Transition to Implementation

Describe the planned next steps for the Implementation Phase. Identify the order in which components will be built and any outstanding design tasks that must be resolved before implementation begins.

[Describe the implementation sequence and any remaining pre-implementation tasks]

11. References

List all sources cited in this document using the institution's prescribed referencing style (typically Harvard or APA 7th edition). Minimum five academic or industry sources are expected for a postgraduate submission.

[Reference 1]

[Reference 2]

[Reference 3]

[Add all references in the prescribed format]

12. Appendices

Include supporting materials that are too detailed for the body of the document but are necessary for a complete understanding of the design.

Appendix A — Additional Diagrams

[Insert any additional diagrams not included in Section 5]

Appendix B — Code Snippets (if applicable)

If prototype code has been developed, include representative code snippets here. All code must be formatted in a monospaced font (e.g. Courier New) with line numbers.

[Insert representative code snippets (prototyping code only, not full implementation)]

Appendix C — Tools and Software Used

Tool / Software	Version	Purpose / Use in Project
[Tool name]	[Version]	[Describe purpose]
[Tool name]	[Version]	[Describe purpose]
[Tool name]	[Version]	[Describe purpose]
[Tool name]	[Version]	[Describe purpose]
[Tool name]	[Version]	[Describe purpose]

SELF-ASSESSMENT CHECKLIST — RUBRIC ALIGNMENT

Complete this checklist before submitting. Each criterion maps directly to the assessment rubric.
Honest self-assessment supports reflection and learning.

Section Reference	Assessment Criterion	Max Marks	Self-Score
Section 4	Design Documents (High-Level & Detailed)	30	[/ 30]
Section 5	Diagrams and Visual Aids	25	[/ 25]
Section 6	Prototypes and Mockups	25	[/ 25]
Section 7	Requirements Traceability	20	[/ 20]
	TOTAL	100	[/ 100]

Criterion	Completed? (Yes / Partial / No)
All required diagrams are clearly labelled with figure numbers and captions	[Yes / Partial / No]
At least three distinct diagram types have been included	[Yes / Partial / No]
Prototype screenshots are annotated and each image is captioned	[Yes / Partial / No]
Traceability matrix links every requirement to a design element	[Yes / Partial / No]
Stream-specific section (Section 8) has been fully completed	[Yes / Partial / No]
Security and risk evaluation has been addressed (Section 9)	[Yes / Partial / No]
All references are formatted consistently in the required citation style	[Yes / Partial / No]
Appendices are complete and referenced in the body text	[Yes / Partial / No]
Document has been proofread and uses formal academic language throughout	[Yes / Partial / No]

— End of Design Phase Submission Template —